Wikontic: Constructing Wikidata-Aligned, Ontology-Aware Knowledge Graphs with Large Language Models

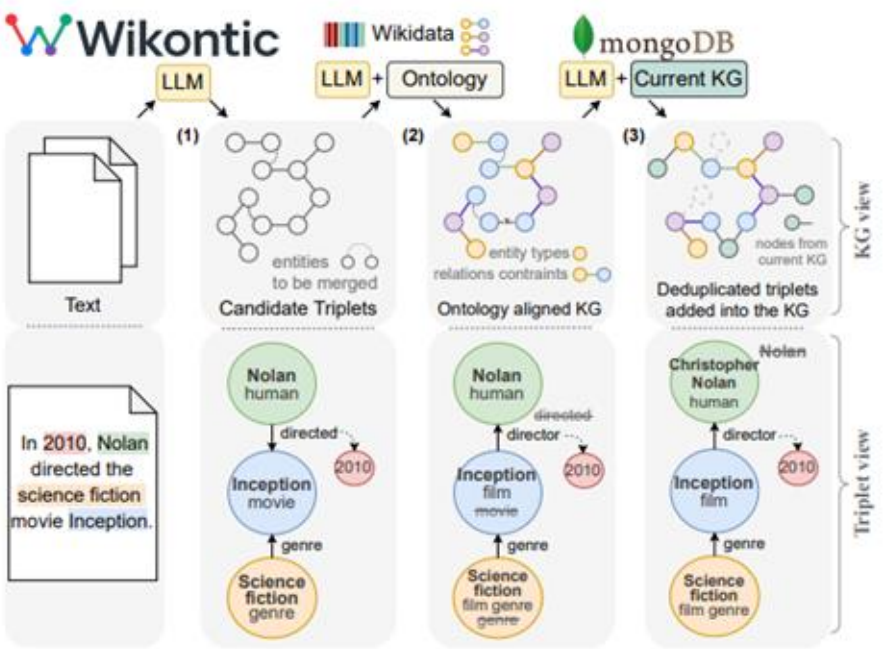
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What is Wikontic?

- Knowledge graphs have commonly been used to organize information from large corpuses into easy-to-retrieve relations, allowing an LLM to find relevant background information in a timely manner
- However, the approach of using knowledge graphs as a middleman between the LLM and the original knowledge base was underexplored, since it was difficult to enforce consistency among node representations and reduce redundancy among relations
    - “NYC located in USA” versus “New York City in-country United States”
- Wikiontic leverages Wikidata information to build a more consistent knowledge graph, and then uses the knowledge graph itself as a primary data source, rather than connecting a relation back to its relevant text source

How is Wikontic constructed?

- Subject, relation, object triplets are extracted from a knowledge base text via LLM
- Using Wikidata, a custom ontology database is created. From this database, an LLM selects the best database relations that match the triplet
- The subject and object names are then compared to existing nodes as the knowledge graph is constructed, and aliased to existing nodes if possible



How does the Wikontic knowledge graph compare to other works?

- Other approaches towards improving knowledge graphs, such as KGGen (Mo et al., 2025), do not have an ontology relational constraint, instead relying on clustering algorithms to reduce the redundancy in the knowledge graph. Wikontic outperformed KGGen on the MINE-1 benchmark, showing that it preserves information better as a result of Wikidata ontologies
- SynthIE (Josifoski et al., 2023) meanwhile uses a knowledge graph generated from Wikidata, and then supplements it with synthetic text based on Wikidata triplets. Wikontic avoids using original text at all during the retrieval phase, so it avoids both the computation needed for that text generation and avoids any hallucinations or missed details that comes with synthetic data
- HOLMES (Panda et al., 2024) builds a knowledge graph by reconstructing it partially based on the prompt, while Wikontic constructs the knowledge graph just once, not needing to recompute the graph each time

How does Wikontic compare to other works in information retrieval?

- Wikontic does not access the original text at any point, which is a fundamental design difference from other RAG systems, such as AriGraph (Anokhin et al., 2025). This significantly reduces the amount of tokens needed, with AriGraph specifically taking up almost 3 times the amount of tokens
- The main weakness of Wikontic's approach is that knowledge graph triplets cannot inherently contain more information that the source text itself
- While its performance is comparable to AriGraph and HOLMES even without the original text, its performance was still overall worse

Method	MuSiQue		HotpotQA	
	EM	F1	EM	F1
Wikontic, gpt4.1	46.8±0.8	**59.8**±0.3	64.5±0.4	76.0±0.4
Wikontic, gpt4.1-mini	42.6±0.7	55.9±0.3	59.7±0.6	71.7±0.8
Wikontic, gpt4o-mini	42.1±0.1	53.3±0.1	53.7±1.2	65.8±1.0
Full context, gpt4	33.5	42.7	53.0	68.4
Supporting facts, gpt4	45.0	56.0	57.0	73.8
ReadAgent (Lee et al., 2024), gpt4	35.0	45.1	48.9	62.0
GraphReader (Li et al., 2024), gpt4	38.0	47.4	55.0	70.0
GraphRAG (Edge et al., 2024), gpt4o-mini	40.0	53.5	58.7	63.3
AriGraph (Anokhin et al., 2025), gpt4o-mini	36.5	47.9	60.0	68.0
AriGraph (Anokhin et al., 2025), gpt4	45.0	57.0	**68.0**	74.7
HOLMES (Panda et al., 2024), gpt4	**48.0**	58.0	66.0	**78.0**
Wikontic, Llama 3.3	**37.7**±0.6	**49.7**±0.4	55.1±0.5	67.4±0.5
HippoRAG v2 (Gutiérrez et al., 2025), Llama 3.3	37.2	48.6	**62.7**	**75.5**

How could we expand upon Wikontic?

- Wikontic's advantages are mainly in the structure of its knowledge graph, while its main weakness is that information retrieval without the original text is inherently limited
- The obvious improvement is to simply leverage the Wikontic approach to building a knowledge graph and combine it with a RAG system that does use the original text
    - HOLMES generates a knowledge graph based on the question asked, and then uses that graph to retrieve the source text. Using a ontological constraint for that knowledge graph could improve the accuracy of retrieved information
    - AriGraph generates an episodic memory graph as it traverses, which could be informed by a Wikidata ontology database while created
- Work could also be done to expand on the graph structure of Wikontic to incorporate more detail into relations in addition to the existing qualifier and sub-entity relationships

Sources

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